

Optimization of Overactive Bladder Medications in Older Adults Residing in Long-Term Care Facilities

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BACKGROUND: Urinary incontinence is a condition prevalent in older adults and has the potential to impact health and quality of life in patients. Commonly prescribed anticholinergic medications have significant side effects that are often heightened in older adults and should be optimized and de-escalated as warranted. Given pharmacists' focus on medication regimens, pharmacists are uniquely equipped to help optimize overactive bladder (OAB) medications in long-term care.

OBJECTIVE: To reduce inappropriate use of OAB medications in older adults residing within long-term care facilities.

METHODS: An intervention-based quality improvement project was completed including 27 patients who were residing at **multiple care settings within the long-term care facilities studied**. A medication review and patient interview were conducted by a clinical pharmacist to assess OAB medication appropriateness. Recommendations for therapy modification were made to the medical team to reduce anticholinergic adverse effects while maintaining treatment effect.

RESULTS: A total of eight patients (57.1%) in the intervention group reported experiencing lessened or eliminated anticholinergic adverse effects compared with only one patient (10%) reporting the same in the non-intervention group ($P = 0.0333$; OR [95% CI] = 10.7 [1.007-587.8]). Effects of intervention on OAB symptoms were variable but most patients did not experience worsening of symptoms. A total of 14 OAB medications were deprescribed or dose-reduced.

CONCLUSION: Pharmacist optimization of OAB medications for older individuals residing in long-term care facilities is associated with elimination of potentially unnecessary medications, potentially with improvement of adverse effects and without worsening of the OAB symptoms.

KEY WORDS: Anticholinergic medications, Deprescribing, Long-term care, Overactive bladder, Pharmacy, Prevention.

ABBREVIATIONS: AE = Adverse effects, CDC = Centers for Disease Control and Prevention, CI = Confidence interval, LECOM = Lake Erie College of Osteopathic Medicine, LNR = LECOM Nursing and Rehabilitation, LVS = LECOM at Village Square, OAB = Overactive bladder, OR = Odds ratio, SLC = LECOM Senior Living Center.

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Introduction

Urinary incontinence is a common condition in older individuals and has the potential to significantly impact health and quality of life. The condition is estimated to affect 423 million adults worldwide¹ and prevalence is greater among older adults. According to the most recent statistics released by the Centers for Disease Control and Prevention (CDC), an estimated 43.8% of noninstitutionalized patients older than 65 years of age and 49.6% of nursing facility residents older than 65 years of age have some degree of urinary incontinence.² While there are various types of incontinence, one of the most often treated with medications is urge incontinence, which is usually associated with overactive bladder (OAB). Whether or not OAB causes incontinence in a patient, the pharmacotherapies are the same. Patients are prescribed either an anticholinergic medication, such as oxybutynin, or a beta₃-adrenergic agonist, such as mirabegron and vibegron, to control OAB symptoms.

While OAB medications show modest efficacy and improve symptoms when combined with behavioral therapy,^{3,4} the commonly prescribed anticholinergic medications have significant adverse effects that are often increased in older adults. Potential short-term adverse effects (AEs) of anticholinergic medications may include dry mouth, constipation, blurred vision, tachycardia, drowsiness, and decreased cognitive function. Prolonged and cumulative dose exposure may lead to cognitive impairment.⁵ Some patients also take other anticholinergic medications such as antipsychotics, antispasmodics, and first-generation antihistamines, which increase the anticholinergic burden on these patients and therefore worsen both short- and long-term AEs. Novel beta₃-adrenergic agonists, which are known to have more mild AE profiles compared with the anticholinergics, are considered by experts to still be only modestly effective, expensive compared with alternative agents, and have additional AEs, such as mirabegron's blood pressure increases.⁶ These medications should be used with caution, optimized and de-escalated as warranted, and used in combination with behavioral measures to improve patient's OAB and urinary incontinence.

Another aspect which contributes to the AEs inherent to using OAB medications is the high rate of polypharmacy reported in older adults. Polypharmacy, defined as a patient taking 5 or more medications, is estimated to be as high as 65% for adults 65 years of age or older in the United States.⁷ In long-term

care facilities, this problem is even more significant with the use of 5, 9, and 10 medications being reported in up to 91%, 74%, and 65% of residents internationally, respectively.⁸ Polypharmacy has negative consequences such as increased risk of AEs and events, inappropriate medication use, prescribing cascades, and increased mortality. One way to reduce or prevent polypharmacy is deprescribing, a process where inappropriate medications are discontinued, and medication dosages are reduced to minimum effective doses to optimize clinical outcomes.

Pharmacists are in a unique position to aid in the optimization and de-escalation of OAB medications in this patient population. Given pharmacists' focus on medication therapy management, they are equipped to recognize potential drug interactions, which could contribute to OAB and the AEs of OAB medications. There is limited literature about pharmacist-based de-escalation of urinary anticholinergic medications; however, prior studies have shown that such a program is feasible in community-dwelling older adults, even when benefit to cognitive and physical functioning is not clear currently.⁹

Design

This study was conducted in three phases: pre-implementation, implementation, and post-implementation. The pre-implementation phase occurred between December 18, 2023, and February 8, 2024, where patients were screened for eligibility and interviewed. The implementation phase occurred between February 9, 2024, and April 9, 2024, where recommendations were provided to attending physicians to optimize OAB medications. Finally, the post-implementation review occurred between April 10, 2024, and April 16, 2024, where patients were interviewed a second time to determine the outcome of medication optimization.

Setting

This study was completed at three skilled nursing facilities.

Patients, Participants

Residents were considered for inclusion if they were 50 years or older and were prescribed OAB medication. Individuals were excluded if they were documented hospice patients, short-term rehabilitation patients, or consistently following with a urologist. Out of



40 patients screened, 27 were included in the study and were assessed for potential intervention. Out of those patients, 24 remained admitted to the facilities and were available to finish the study.

Interventions

Baseline demographics were collected via electronic medical record (EMR) for each patient enrolled in the study. Data collected included age, gender, serum creatinine, creatinine clearance, OAB medications, and any documentation of anticholinergic AEs. Patients who were eligible for enrollment were then interviewed during the pre-implementation phase using a patient questionnaire. The questionnaire collected the following data: number of urges to void per day, number of incontinence episodes per day, number of episodes of urination per day (regardless of incontinence), number of nocturnal urinations per night, and any anticholinergic AEs. If a patient was cognitively impaired, but otherwise eligible, the interviewing pharmacist spoke with nursing staff instead. Patients and nursing staff were re-interviewed during the post-implementation phase to determine the primary and secondary outcomes. The primary outcome of this study was improvement of anticholinergic AEs with secondary outcomes including effect on symptoms of OAB and change in each anticholinergic AE.

A medication review from a clinical pharmacist was completed for all participants to assess for appropriateness of their OAB medications. Once the medication review was completed, an interview was conducted with either the patient or nursing staff based on cognitive impairment. Following the interview, the clinical pharmacist developed an intervention, or lack thereof, based on information gathered and clinical judgment. Recommendations for medication changes were communicated to the medical team managing the patient. To limit variables and maximize potential for intervention acceptance, interventions only involved OAB medications, with no adjustment to contributing medications (eg, diuretics), anticholinergic medications, or initiation of non-pharmacological methods. Providers were aware of this project when reviewing recommendations. All pharmacy recommendations were accepted by the medical team and implemented. Interventions were categorized into changing an anticholinergic medication to a beta₃-adrenergic agonist, increasing the dose of a beta₃-adrenergic agonist, decreasing the dose of the OAB medication, and discontinuation of the OAB medication.

Two months after the intervention, the clinical pharmacist completed the post-implementation interview as a follow-up.

Statistical Analysis

Statistical analyses were performed on baseline characteristics to rule out confounding bias with an unpaired *t*-test for patient age, serum creatinine, and creatinine clearance, and a Fisher's exact test for gender.

Results

Out of 40 patients screened, 31 (77.5%) patients met eligibility criteria for inclusion into the study. Of these 31, 4 declined to be interviewed, with 27 (67.5%) entering the implementation phase. Interventions were recommended for 16 (59.3%) of the 27 who moved on to this phase. Three patients (2 in the intervention group, 1 in the non-intervention group) were lost to follow-up during the implementation phase leading to only 24 (60%) patients available for the post-implementation interview. Table 1 shows the baseline characteristics of the participants. Table 2 shows the primary outcome (improvement of anticholinergic AEs) and effect of intervention on OAB symptoms. Table 3 shows number of patients reporting each anticholinergic AE before and after intervention. Figure 1 shows the breakdown of interventions performed. Figures 2 through 5 show the comparison pre- and post-intervention of each urinary measure.

The mean age of patients was 75.8 years in the intervention group and 78.4 in the non-intervention group. Most patients were female with a mean creatinine clearance of 69.2 mL/min, indicating mild kidney impairment. At baseline, 18 patients were taking an anticholinergic medication, with 11 taking a beta₃-adrenergic agonist; 2 of these patients were taking both. There were no statistically significant differences for any baseline characteristics between groups.

The most common intervention completed was switching the anticholinergic medication to a beta₃-adrenergic agonist based on the lower rate of AEs. This change was completed in 6 patients, followed by discontinuation of medication in 5 patients. When including the 3 patients who had their OAB medication dose reduced, a total of 14 medications were deprescribed or dose reduced. The 2 intervention group patients who were lost to follow-up had their OAB medication discontinued.



Table 1. Baseline Characteristics (n = 27)

Characteristic	Intervention Group (n = 16)	No Intervention Group (n = 11)	P-Value
Age (Years)	75.8	78.4	0.60
Female (%)	75	77	1
SCr (mg/dL)	1.26	0.99	0.47
CrCl (mL/min)	64.81	75.61	0.60
Lost to Follow-up	2	1	N/A
Overactive Bladder Agent			
Anticholinergic	13	5	N/A
Beta ₃ -Agonist	4	7	N/A

Abbreviations: CrCl = Creatinine clearance, SCr = Serum creatinine.

Table 2. Improvement of Anticholinergic Adverse Effects and Effect of Intervention on OAB Symptoms (n = 24)

OAB Effect	Change Medication (n = 6)	Decrease Dose (n = 3)	Discontinue (n = 3)	Increase Dose (n = 2)	Total Interventions (n = 14)	No Intervention (n = 10)
Improved Side Effects	4	2	1	1	8	1
Positive Effect on OAB	3	0	0	1	4	0
Neutral Effect on OAB	2	2	1	1	6	9
Negative Effect on OAB	1	1	2	0	4	1

Abbreviation: OAB = Overactive bladder.

During the post-implementation interviews, 8 patients (57.1%) in the intervention group reported reduced or no anticholinergic AE, compared with only 1 patient (10%) reporting the same in the non-intervention group. As for effects on OAB symptoms, of the intervention group, 4 patients (28.6%) reported a reduction of symptoms, 4 patients (28.6%) reported a worsening of symptoms, and 6 patients (42.8%) reported no effect on their symptoms. This is contrasted with the 9 patients (90%) reporting no change in symptoms in the non-intervention group and 1 patient (10%) reporting worsening symptoms.

Per Table 3, individual anticholinergic AEs were reported less in the intervention group post-implementation except for one patient who was reported to have more falls after discontinuation of their OAB medication. Change in patient reports of dizziness (4 patients vs 1 patient), xerostomia (9 patients vs 6 patients), and constipation (7 patients vs 4 patients) were the most frequent changes in the intervention group post-implementation. Number of urges per day changed from 4.86 to 4.36, incontinence episodes per day changed from 3.96 to 3.43, episodes of urination regardless of urge per day changed from 5.50 to 5.21, and number of nocturnal urinations per night were reduced from 1.57 to 1.46.



Table 3. Reported Anticholinergic Adverse Effects by Intervention (n = 24)

Adverse Effects	Change Medication (n = 6)	Decrease Dose (n = 3)	Discontinue (n = 3)	Increase Dose (n = 2)	Total Interventions (n = 14)	No Intervention (n = 10)
Pre-Intervention Dizziness	2	1	0	1	4	2
Post-Intervention Dizziness	0	1	0	0	1	1
Pre-Intervention Falls	0	0	0	0	0	1
Post-Intervention Falls	0	0	1	0	1	1
Pre-Intervention Fatigue	4	2	0	1	7	3
Post-Intervention Fatigue	2	1	1	1	5	2
Pre-Intervention Sedation	0	0	0	0	0	1
Post-Intervention Sedation	0	0	0	0	0	1
Pre-Intervention Xerostomia	5	3	1	0	9	5
Post-Intervention Xerostomia	4	1	0	1	6	5
Pre-Intervention Constipation	3	3	1	0	7	2
Post-Intervention Constipation	2	1	1	0	4	1

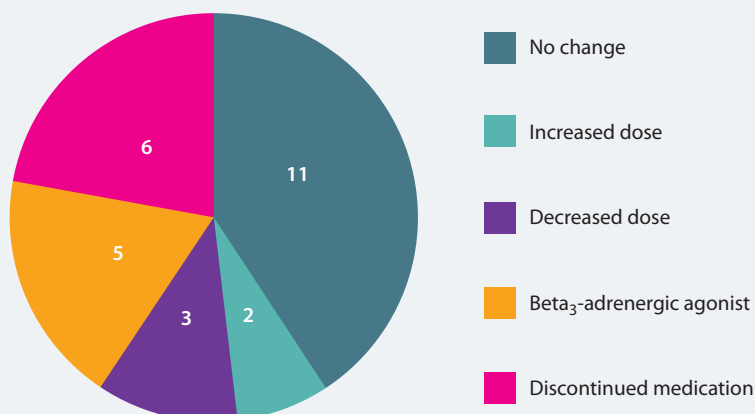
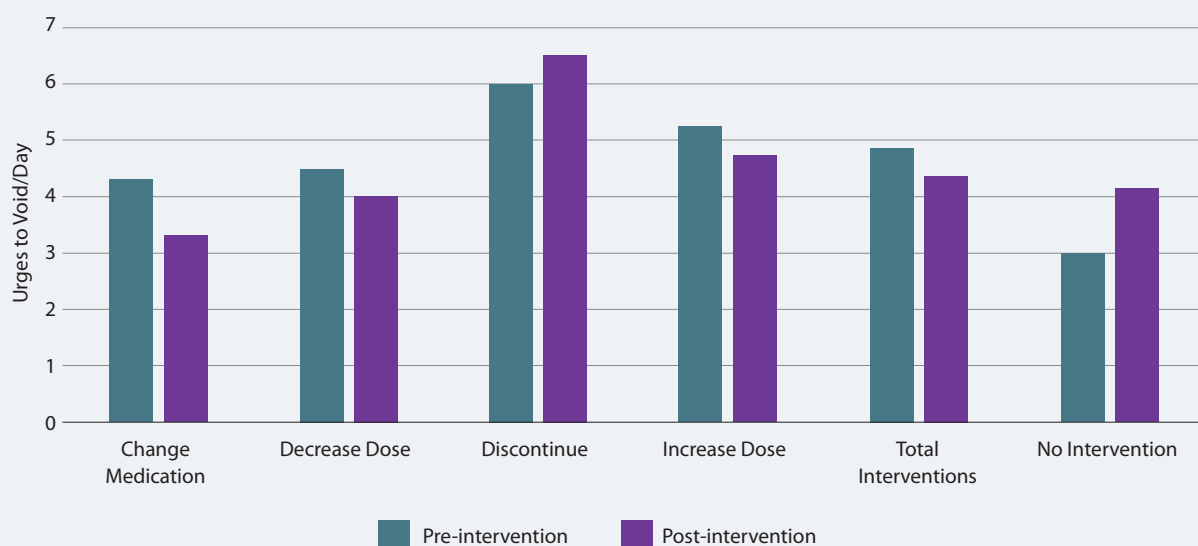
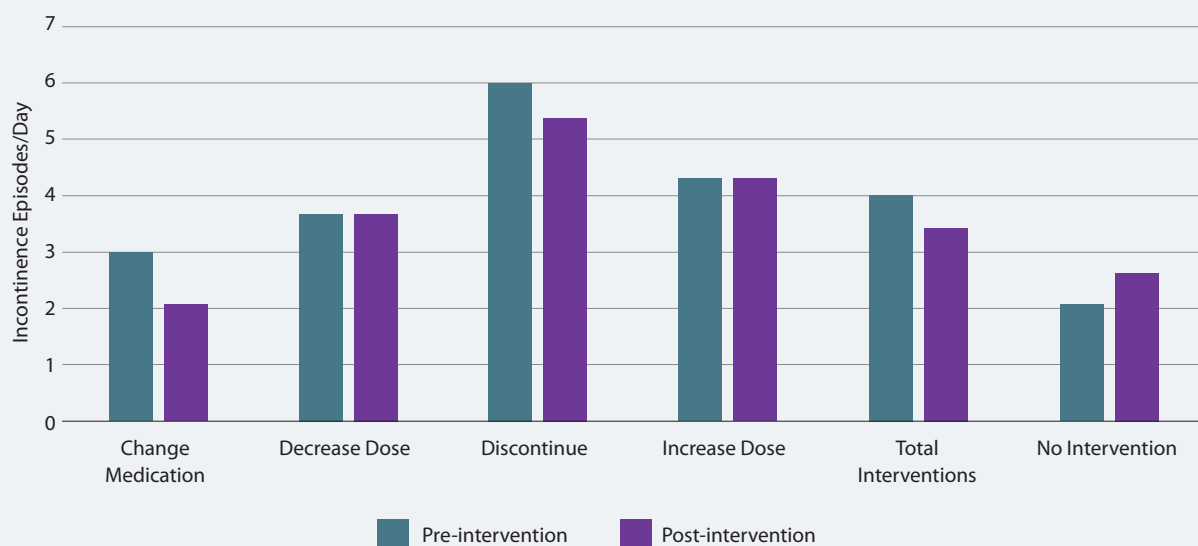
Figure 1. Interventions Made

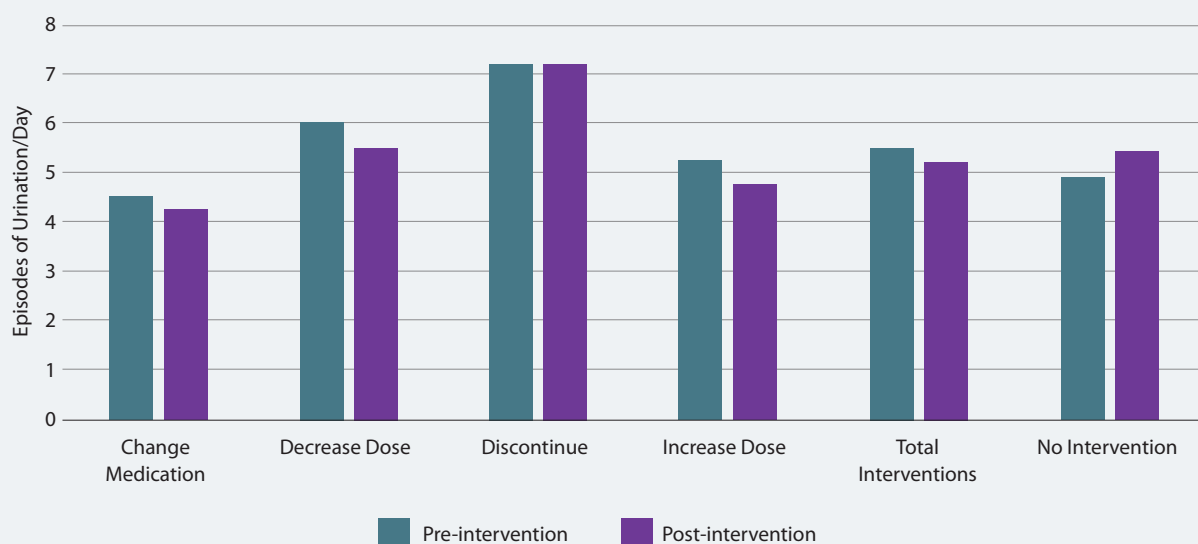
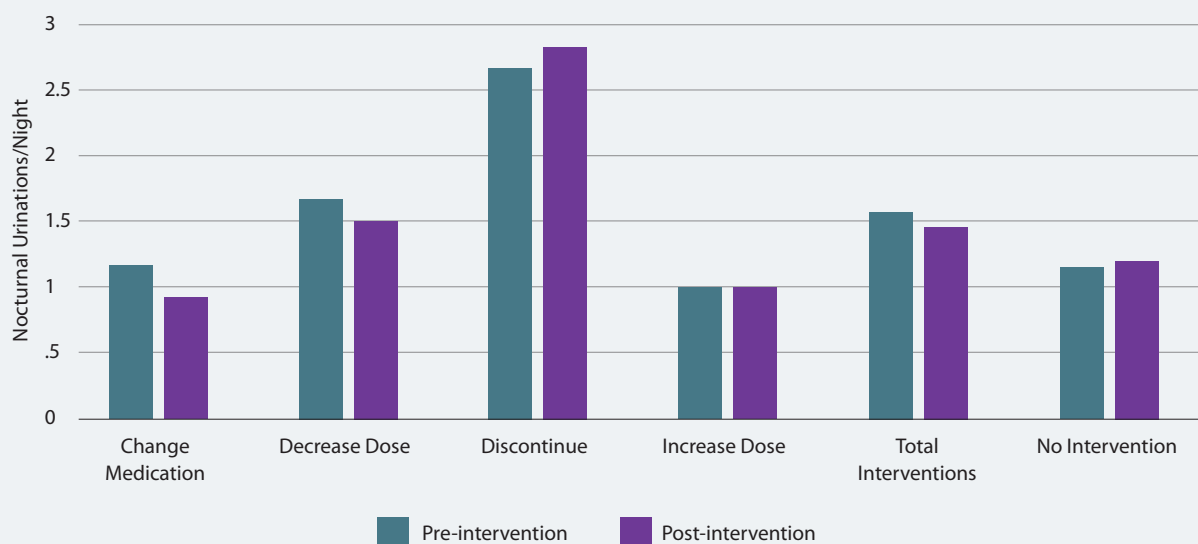
Figure 2. Number of Urges to Void/Day by Intervention**Figure 3. Incontinence Episodes/Day by Intervention**

Discussion

The goal of this study was to reduce the overuse of OAB medications in older adults residing within the long-term care facilities studied and to demonstrate that pharmacists are uniquely equipped to aid in the optimization and de-escalation of OAB medications in the older population. The results show that, by using medication interventions, we were able to

deprescribe anticholinergic medications in eight patients and reduced or eliminated anticholinergic AEs in doing so. While the effects on OAB symptoms were variable according to the results, the effects were in line with the interventions made. Of note, two patients whose OAB medications were discontinued because of cognitive impairment and complete urinary incontinence reported more frequent urination with one patient experiencing increased falls due



Figure 4. Episodes of Urination/Day by Intervention**Figure 5. Nocturnal Urination/Night by Intervention**

to increased bathroom use. While this may indicate optimization as a complicated and patient-specific process. Overall, these interventions reportedly had a positive effect on patient quality of life.

There are several limitations of this study that may affect interpretation of the results. While other anticholinergic medications were monitored in case of change during the intervention phase, overall

anticholinergic burden was not assessed. While no changes to other anticholinergic medications occurred during the study period, the lack of this assessment represents a potential impact on the lack of reporting for anticholinergic AEs. Many of the measures gathered for urinary frequency are subject to the memory of the patient or the health care worker interviewed, which may potentiate recall bias. In addition, six patients were cognitively impaired to the point of requiring



the interview portions to be conducted with nursing staff. Given the timeline needed to complete this project, difficulties with reaching specific nursing staff, and nursing staff turnover, we were unable to obtain interviews with the same nurses for each interview. This introduces another source of bias beyond the bias inherent in these proxy reports. The study was not blinded; therefore, it was subject to performance bias as patients who received an intervention may have reported improvement because of expecting improvement from an intervention. Response bias is also a reasonable worry as patient may not have given truthful answers to questions regarding the topic of urination and incontinence because of embarrassment. The use of a questionnaire designed specifically for this project, which is a non-validated instrument, introduces potential for information bias to the results. Finally, the small sample size and short duration of follow-up posed as a limitation for this review.

Conclusion

Pharmacist optimization of OAB regimens for older adults in long-term care is associated with deprescription of potentially unnecessary medications with the potential for urinary symptoms to not worsen. A total of eight patients reported improvement of AEs in the intervention group, which demonstrates additional benefits of considering dose reductions or discontinuing this class of medications. These results will be used to support ongoing pharmacy efforts to further deprescribe OAB medications when appropriate, in order to reduce polypharmacy and the risk of anticholinergic AEs in older adults. Implications for future clinical research and practice include continued optimization of OAB medications, adjustment of medications contributing to OAB (such as diuretics), and introduction of non-pharmacological methods of controlling OAB symptoms.

REFERENCES

1. Irwin DE, Kopp ZS, Agatep B, et al. Worldwide prevalence estimates of lower urinary tract symptoms, overactive bladder, urinary incontinence and bladder outlet obstruction. *BJU Int*. 2011;108(7):1132-8.
2. Gorina Y, Schappert S, Bercovitz A, et al. Prevalence of incontinence among older Americans. *Vital Health Stat 3*. 2014;(36):1-33.
3. Burgio KL, Kraus SR, Johnson TM, et al. Effectiveness of combined behavioral and drug therapy for overactive bladder symptoms in men: a randomized clinical trial. *JAMA Intern Med*. 2020;180(3):411-419. doi:10.1001/jamainternmed.2019.6398.
4. Balk EM, Rofeberg VN, Adam GP, et al. Pharmacologic and nonpharmacologic treatments for urinary incontinence in women: a systematic review and network meta-analysis of clinical outcomes. *Ann Intern Med*. 2019;170(7):465-79. doi:10.7326/M18-3227.
5. Gray SL, Anderson ML, Dublin S, et al. Cumulative use of strong anticholinergics and incident dementia: a prospective cohort study. *JAMA Intern Med*. 2015;175(3):401-7. doi:10.1001/jamainternmed.2014.7663.
6. Sebastianelli A, Russo GI, Kaplan SA, et al. Systematic review and meta-analysis on the efficacy and tolerability of mirabegron for the treatment of storage lower urinary tract symptoms/overactive bladder: comparison with placebo and tolterodine. *Int J Urol*. 2018;25(3):196-205. doi:10.1111/iju.13498.
7. Young EH, Pan S, Yap AG, et al. Polypharmacy prevalence in older adults seen in United States physician offices from 2009 to 2016. *PLoS One*. 2021;16(8):e0255642. Published August 3, 2021. doi:10.1371/journal.pone.0255642.
8. Jokanovic N, Tan EC, Dooley MJ, et al. Prevalence and factors associated with polypharmacy in long-term care facilities: a systematic review. *J Am Med Dir Assoc*. 2015;16(6). doi:10.1016/j.jamda.2015.03.003.
9. Ha M, Furman A, Al Rihani SB, et al. Pharmacist-driven interventions to de-escalate urinary antimuscarinics in the Programs of All-Inclusive Care for the Elderly. *J Am Geriatr Soc*. 2022;70(11):3230-8. doi:10.1111/jgs.17965.

